

Dimensions

What are Dimensions of a Shape

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The dimension of an object/shape is basically a topological measure of the length, width, or height of its covering properties. Dimension is a measure of size extended in a specific direction. In simple terms, it is the number of coordinates required to specify a point on the object. For instance, a **rectangle** is a 2-dimensional figure, while a **cube** is 3-dimensional. The letter D in 2D, 3D denotes dimensions. The dimension of an object is sometimes also called its "dimensionality".

(Image will be uploaded soon)

Types of Dimensions

Most commonly, there are 4 types of dimensions as given below:

- 1. Zero Dimension:** A point consists of no extent in any direction, so it has no dimension.
- 2. One Dimension:** A line has how many dimensions? A line has only one dimension, which is its length.
- 3. Two-Dimension:** A piece of paper consists of two dimensions i.e. length and width.
- 4. Three-Dimension:** Dices consist of three dimensions i.e., length, width, and height. Dimension implies the measurement of the size or distance of an object or space in only one direction.

Formula of Dimension

How many dimensions does a point have? Dimension is a measure of a line or point stretched out in one direction and we are aware that every shape around us consists of some dimensions. The theory of dimension does not contain any

particular dimension formula. The value and the number of dimensions differ depending upon their size and shape. The following table depicts the dimension formulas of circles and polygons.

Dimension Formulas

Number of Dimensions	Physical Quantity	Dimension Formula	Dimensions Used
2D - Polygons	Area	length $\times$ breadth (or) height	length, breadth (or) height
	Perimeter	Adding up all the distance around the object or shape	distance
2D - Circle	Area	πr^2 (or) πd	radius (or) diameter
	Circumference	$2 \pi r$	radius

Dimensions can be used to identify the unknown values of shapes using the help of formulas. For instance, if the perimeter of a [square](#) is 12 inches, we can determine its length with the help of the dimension formula: Perimeter = 6 $\times$ Length. Substituting the value of perimeter in the formula, we obtain 12 = 6 $\times$ Length. Thus, Length = 2 inches.

Different units are used to measure different quantities like length, temperature, area etc. The following table shows the dimensions that are usually used for units like length, area, temperature, volume and weight.

Dimensions of Units

Quantity	Units
Length	inches, feet, yards, miles
Area	Square Inch, Square Feet, Square Yard, Square Mile, Acre

Temperature	Celsius, Fahrenheit, Kelvin
Volume (Capacity)	quarts (qt), pint (pt), gallons (gal), fluid ounces (fl oz)

How Many Dimensions Does a Plane Have

Planes are 2 dimensional. However, they can exist in 3-dimensional spaces as well. The plane contains two dimensions: length and width. But since the plane is limitlessly large, the length and width are unable to be measured. Just as a line is defined as a 1-dimensional figure or one-dimensional shape having two points, a plane is defined by three points.

Dimensional Analysis

Dimensional analysis means to determine the physical quantities and evaluate the association between them. It helps us to study the nature of different objects like straightness and flatness mathematically.

Dimensionless Quantities

The quantities that contain no dimension are known as dimensionless quantities. Such quantities consist of a certain value. For instance, the constant π is a dimensionless quantity since it has a fixed value. Following are the types of Dimensionless quantities:

- 1. Dimensionless Quantities With Units:** These include angular displacement – Joule's constant – joule/calorie, radian etc.
- 2. Dimensionless Quantities Without Units:** These include pure [numbers](#), π , $\sin \theta$, $\cos \theta$, $\tan \theta$, e , etc.

Dimensional Formula

The expression displaying the powers to which the fundamental units are to be raised in order to get one unit of a derived quantity is known as the dimensional formula of that quantity.

If Q is the unit of a derived quantity denoted by the letter $Q = M^a L^b T^c$, then $M^a L^b T^c$ is known as the dimensional formula and the [exponents](#) a , b and, c , are what we call the dimensions.

Fun Facts

- It is believed that the Universe consists of 10 dimensions; however, fortunately, the human eye can only apprehend three.
- The meaning of dimension differs in mathematics and physics.
- Dimension is the point or value, measuring the object.

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